

Metaheuristic-Optimized Intrusion Detection: A Comparative Study of GA, PSO, and SA on CICIDS2017

Ziad Ahmed Belal Ibrahim Alei Halawa Mohammed Senan

School of Computer Science, University of Nottingham Malaysia

Semenyih, Selangor, Malaysia

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Abstract Intrusion Detection Systems (IDS) face two persistent challenges: high-dimensional feature spaces and poorly tuned model parameters. This paper presents a comparative study of three metaheuristic algorithms, Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Simulated Annealing (SA), applied to jointly optimize feature selection and Random Forest hyperparameters on the full four-day CICIDS2017 dataset (2,300,825 records, 14 attack types). All algorithms successfully reduced the feature set while maintaining strong detection performance above a 98.9% F1 score. SA achieved the best overall results with a 99.5879% F1-score and 0.1956% FPR using 26 features, followed by GA with a 99.2083% F1-score using only 9 features. PSO, while fastest to converge, showed the weakest performance on the multi-day dataset with a 98.9581% F1-score, revealing sensitivity to dataset complexity. These results demonstrate that metaheuristic optimization is an effective and practical approach to improving IDS performance in realistic network environments.

Keywords: Intrusion Detection System, Metaheuristic Optimization, Feature Selection, Genetic Algorithm, Particle Swarm Optimization, Simulated Annealing, Random Forest, CICIDS2017

I. INTRODUCTION

The rapid growth of network systems and the increasing number of connected devices have made cybersecurity a major concern today. Intrusion Detection Systems (IDS) are essential tools that help protect networks by analyzing traffic and detecting suspicious activities that may indicate an attack.

However, traditional IDS approaches often face a challenge in balancing detection performance and computational cost. Network datasets usually contain a large number of features, and many of these features can be redundant, unnecessary, or only weakly related to the classification task. Using all features can increase processing time and sometimes even reduce performance due to the curse of dimensionality [1].

Machine learning has become a key solution in modern IDS. Models such as Random Forest, Support Vector Machines, and gradient boosting have shown strong performance on datasets like CICIDS2017, NSL-KDD, and UNSW-NB15. However, their performance depends heavily on two factors: the selected features and the hyperparameter settings. Default hyperparameters are not always suitable for every dataset, and traditional tuning methods like grid search become very expensive when the number of combinations increases.

Metaheuristic optimization provides an effective way to solve both problems at the same time. Instead of treating feature selection and hyperparameter tuning separately, they are combined into one optimization problem. This allows the model to improve performance while also reducing the number of features. Previous research has shown strong results using different metaheuristics. Genetic Algorithms (GA) have been effective for feature selection [2], [3]. Particle Swarm Optimization (PSO) has been successful in tuning models like Random Forest [4]. Simulated Annealing (SA) offers a different approach that helps avoid getting stuck in poor local solutions [5].

This paper presents a comparison of these three algorithms when applied to a Random Forest-based IDS using the full four-day CICIDS2017 dataset. The main contributions of this study are (1) building a full preprocessing pipeline using SMOTE with stratified capping; (2) establishing a baseline Random Forest model on the multi-day dataset; (3) applying GA, PSO, and SA for joint feature selection and hyperparameter optimization; and (4) comparing

results using accuracy, precision, recall, F1-score, and false positive rate (FPR).

The remainder of this paper is organized as follows. Section II explains the methodology. Section III presents the results. Section IV discusses the findings, and Section V concludes the paper.

II. METHODOLOGY

A. Dataset

This study uses the CICIDS2017 dataset across four days of traffic capture: Tuesday, Wednesday, Thursday, and Friday. Monday was excluded because it only contains normal traffic without attacks.

The combined dataset includes 2,300,825 records covering 14 different attack types, such as DoS, DDoS, PortScan, Botnet, and Web Attacks. After cleaning, 2,018,473 records remained. Using multiple days ensures the model is exposed to a wider variety of attack types, making the results more realistic compared to single-day evaluations [6].

For this study, the problem was simplified into binary classification: BENIGN (class 0) and ATTACK (class 1). Before applying SMOTE, there were 1,274,185 benign samples and 340,593 attack samples. After SMOTE, both classes were balanced to 1,274,185 samples each. To reduce computation, the dataset was capped to 400,000 training samples (200,000 per class). The test set contains 403,695 samples and was kept unchanged to ensure fair evaluation.

B. Preprocessing Pipeline

The preprocessing pipeline includes four main steps. First, data cleaning: rows with infinite values, missing values, and duplicate records were removed to avoid bias and data leakage. Second, feature encoding: the label was converted to binary (0 = benign, 1 = attack). Third, normalization: all numerical features were scaled to [0, 1] using Min-Max scaling. Fourth, class balancing: SMOTE was applied to balance the classes, followed by a stratified cap at 400,000 training samples for computational feasibility.

C. Baseline Model

A Random Forest classifier was selected as the base model. It was chosen because it is robust to irrelevant features, provides built-in feature importance scores, and does not require feature scaling [1], [7]. The baseline uses scikit-learn's RandomForestClassifier with default settings ($n_estimators=100$, $max_depth=None$, $min_samples_split=2$, $min_samples_leaf=1$, $max_features='sqrt'$) trained on all 78 features with an 80/20 stratified split.

Table I shows the baseline results on the full four-day dataset.

TABLE I

Baseline Random Forest Performance: CICIDS2017 (All 4 Days)

Metric	Value
Accuracy	99.8514%
Precision	99.5453%
Recall (TPR)	99.7510%
F1-Score	99.6480%
False Positive Rate	0.1218%
Features Used	78
Training Samples	400,000
Test Samples	403,695
TP	84,936
TN	318,159
FP	388
FN	212

The baseline FPR of 0.1218% is higher than what is typically observed on single-day subsets, reflecting the increased difficulty of classifying 14 diverse attack types across all four days.

D. Optimization Framework

The optimization problem is defined as follows. Let $F = \{f_1, \dots, f_n\}$ be the full set of n features and $H = \{h_1, \dots, h_k\}$ be the Random Forest hyperparameters. A candidate solution s consists of a binary feature mask $s_F \in \{0,1\}^n$ and a hyperparameter configuration $s_H \in H$. The goal is to find s^* that maximizes:

$$\text{fitness}(s) = 0.9 \times \text{F1-score}(s) - 0.1 \times (|s_F| / n)$$

This formula rewards a high F1 score while penalizing solutions that use too many features. To reduce computational cost, each fitness evaluation is performed on a 20% stratified sample of the training data. Final evaluation is conducted on complete training and test sets.

E. Algorithm Pseudocode

The three optimization algorithms are summarized below.

Algorithm 1: Genetic Algorithm

```

Initialize N=20 chromosomes (binary feature mask + hyperparams)
FOR each generation (max 30):
    Evaluate fitness for all chromosomes
    IF no improvement for 5 generations: STOP
    SELECT top chromosomes via tournament selection
    CROSSOVER + MUTATE to produce offspring
    KEEP best chromosome (elitism)
EVALUATE best solution on full dataset
RETURN best features, hyperparams, and metrics

```

Algorithm 2: Particle Swarm Optimization

```

Initialize N=20 particles with random positions + velocities
FOR each iteration (max 30):
    SET inertia  $w = 0.9 - (0.5 \times t / \text{MAX\_ITER})$ 
    FOR each particle: evaluate fitness, update pbest, gbest
    IF no improvement for 5 iterations: STOP
    UPDATE velocity:  $v = w \times v + 1.5 \times r1 \times (pbest - x) + 1.5 \times r2 \times (gbest - x)$ 
    UPDATE position:  $x = x + v$ , CLIP to valid range
EVALUATE gbest on full dataset
RETURN best features, hyperparams, and metrics

```

Algorithm 3: Simulated Annealing

```

Initialize random solution s, T=1.0, cooling=0.995
FOR each iteration (max 800):
    GENERATE neighbour s' (flip bits, adjust hyperparams)
    delta = fitness(s') - fitness(s)
    IF delta > 0: ACCEPT s'
    ELSE: ACCEPT s' with probability  $e^{(\text{delta}/T)}$ 
    UPDATE T = T * 0.995
    IF T < 0.001: STOP
EVALUATE best solution on full dataset
RETURN best features, hyperparams, and metrics

```

F. Evaluation Metrics

All models are evaluated using accuracy, precision, recall (TPR), F1-score, false positive rate (FPR), and number of features selected. The F1-score is the primary metric due to class imbalance. FPR is critical for operational usability as it measures false alarms on benign traffic.

III. EXPERIMENTAL RESULTS

A. Baseline Model Performance

Before applying any optimization, a Random Forest model was trained using all 78 features with default hyperparameters on the full four-day CICIDS2017 dataset. The model achieved an accuracy of 99.8514%, an F1-score of 99.6480%, and a false positive rate of 0.1218%. The slightly higher FPR compared to single-day experiments confirms that the multi-day classification task is more challenging, as it includes 14 different attack types. These results serve as the benchmark for evaluating all three metaheuristic algorithms.

Confusion Matrix — Baseline Random Forest (CICIDS2017)

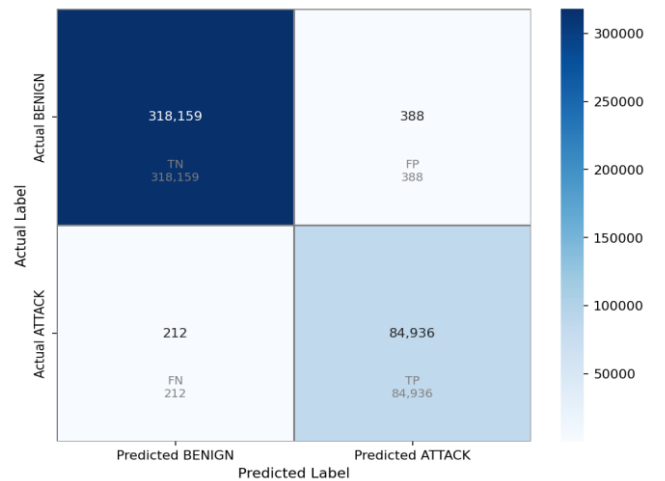


Fig. 1. Confusion matrix of the baseline Random Forest model.

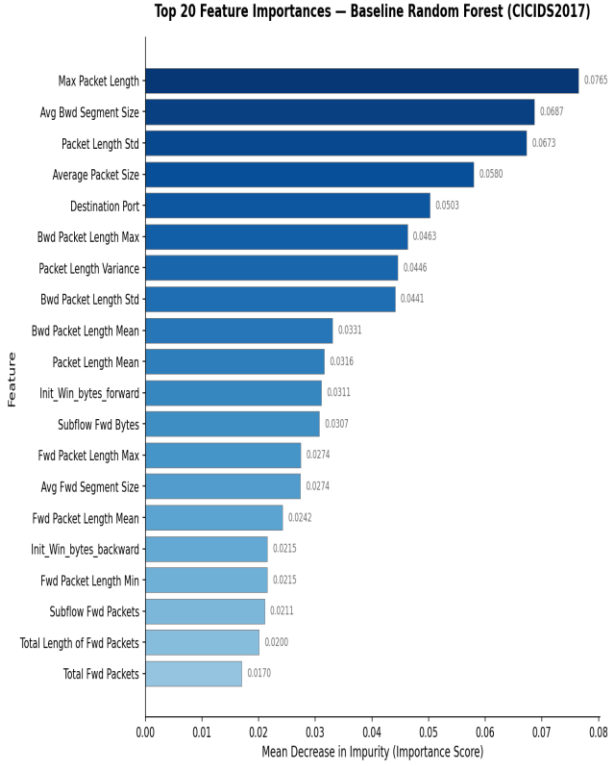


Fig. 2. Top 20 feature importances from the baseline Random Forest model.

B. Genetic Algorithm Results

The GA ran for 18 generations before early stopping was triggered. It selected 9 features from the original 78, achieving the highest feature reduction of 88.5% among all tested algorithms. The optimized model achieved an F1 score of 99.2083% and a FPR of 0.2285%, with a total runtime of 53.2 minutes.

GA improved steadily and reached its best fitness score of 0.88117 at generation 18.

Of the 9 selected features, 4 overlapped with the baseline top 20. The selected features include Destination Port, Flow Duration, Total Backward Packets, Fwd IAT Std, Min Packet Length, Packet Length Mean, URG Flag Count, Subflow Fwd Packets, and Subflow Fwd Bytes. The shorter runtime compared to preliminary Wednesday-only experiments (53.2 vs 575 minutes) is due to the 400,000 sample training cap.

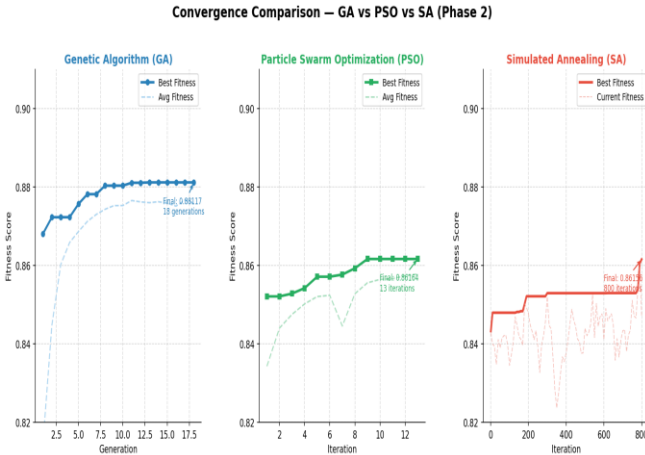


Fig. 3. Convergence comparison of GA, PSO, and SA across iterations.

C. Particle Swarm Optimization Results

PSO ran 13 iterations before early stopping. It selected 24 features (69.2% reduction), achieving an F1-score of 98.9581% and a FPR of 0.4715%, the highest FPR among all models tested. Runtime was 66.7 minutes. As shown in Fig. 3, PSO converged quickly over 13 iterations, reaching a final fitness score of 0.86164. Of the 24 features selected, 5 overlapped with the baseline top 20, mostly related to packet length and inter-arrival time metrics. PSO showed the weakest overall performance on the full multi-day dataset.

D. Simulated Annealing Results

SA ran for the full 800 iterations and selected 26 features (66.7% reduction). It achieved an F1 score of 99.5879% and a FPR of 0.1956%, which is the closest to the baseline FPR among all three algorithms. Runtime was 275.7 minutes. As shown in Fig. 3, SA showed an irregular convergence pattern with fitness values fluctuating due to the temperature-based acceptance mechanism, while the global best improved gradually to reach a final score of 0.86156. Of the 26 features selected, 9 overlapped with the baseline top 20, the highest overlap among all algorithms. SA delivered the strongest overall performance on the full multi-day dataset.

E. Summary of Results

Table II presents a full comparison of all four configurations. Fig. 4 provides a visual comparison of the F1 score, FPR, and features selected. Fig. 5 illustrates the degree of feature reduction relative to the baseline. All three algorithms successfully reduced the feature set while maintaining competitive performance above 98.9% F1.

TABLE II
Master Performance Comparison of all Algorithms

Algorithm	Features	F1-Score	FPR	Runtime
Baseline	78	99.6480%	0.1218%	N/A
GA	9	99.2083%	0.2285%	53.2 min
PSO	24	98.9581%	0.4715%	66.7 min
SA	26	99.5879%	0.1956%	275.7 min

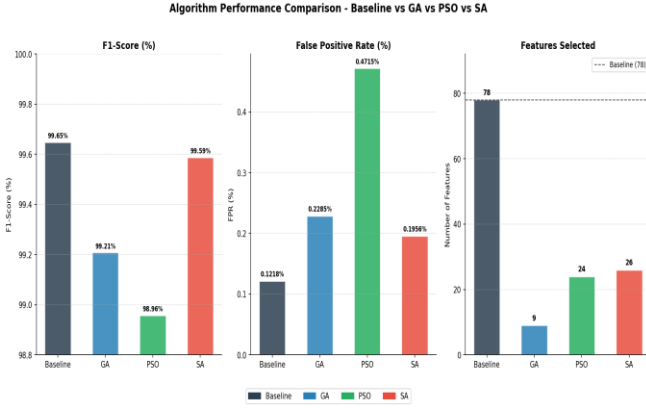


Fig. 4. Performance comparison of algorithms in terms of F1-score, false positive rate, and features selected.

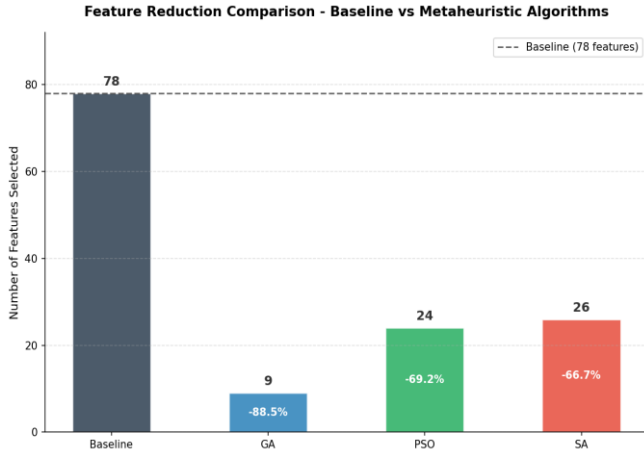


Fig. 5. Feature reduction achieved by each algorithm relative to the baseline.

TABLE III
Feature Overlap Analysis

Algorithm	Features Selected	Overlap / Top 20	Low-Importance
GA	9	4	5
PSO	24	5	19
SA	26	9	17

IV. DISCUSSION

A. Overall Performance

All three metaheuristic algorithms successfully reduced the feature count while maintaining strong detection performance above a 98.9% F1-score. The full multi-day dataset introduced significantly greater classification difficulty, evidenced by the higher baseline FPR of 0.1218% versus 0.0558% in preliminary Wednesday-only experiments. This confirms that using all four days provides a more realistic and challenging benchmark. Notably, the performance rankings changed between single-day and multi-day experiments; SA became the best performer, and PSO became the weakest, a reversal that carries important analytical implications.

B. SA: Best Overall Performer

SA produced the best balance of performance and feature reduction on the full dataset. It achieved 99.5879% F1 and 0.1956% FPR using only 26 features, a 66.7% reduction. The FPR is only 0.0738 percentage points above the baseline, making it the closest algorithm to baseline performance.

This result is particularly significant because SA is a single-solution algorithm. The fact that it outperformed both population-based algorithms on this harder dataset strongly suggests that temperature-based exploration is effective when the fitness landscape is complex and irregular, as is the case with 14 diverse attack types across four days. This is supported by Agrawal et al. [5], who highlighted SA's local optima escape capability as a key advantage in complex IDS feature spaces. Prabakar et al. [8] also noted SA's tendency to be slow but highly effective, which matches our observation of 275.7 minutes runtime paired with the strongest results.

C. GA: Most Aggressive Feature Reduction

GA selected only 9 features, an 88.5% reduction, while maintaining 99.2083% F1. The trade-off is a higher FPR of 0.2285%, nearly double the baseline, which means more false alarms in a deployed system. However, in resource-constrained environments such as edge computing nodes or IoT gateways, this extreme feature reduction could enable real-time detection that would otherwise be computationally infeasible. GA also captured 4 of the top 20 baseline features, confirming it identified some of the most discriminative features even within its compact subset. These findings align with Al-Turaiki and Alshahrani [2], who demonstrated similar accuracy-efficiency trade-offs with GA on CICIDS2017.

D. PSO: Weakest on Full Dataset

PSO was the weakest performer, achieving 98.9581% F1 and 0.4715% FPR, the highest FPR among all configurations. This is particularly notable because in preliminary Wednesday-only experiments, PSO was the strongest performer. This reversal is one of the most significant findings in this study.

PSO's swarm convergence mechanism, which relies on particles following the global best, may cause premature convergence when the fitness landscape is highly irregular. With 14 attack types across four days, the search space becomes multimodal, causing the swarm to cluster around a suboptimal solution early. This is consistent with Kareem et al. [1], who noted PSO struggles in high-dimensional multimodal problems. Despite its weaker performance, PSO completed optimization in only 66.7 minutes, making it suitable for frequent model retraining scenarios.

E. Feature Overlap Analysis

Table III shows that all three algorithms selected many features outside the baseline top 20. SA had the highest overlap (9/20), suggesting it captures primary dataset variance most thoroughly. PSO focused heavily on packet length and timing features. The overall pattern confirms that metaheuristic wrapper methods identify feature combinations that work well together, not just individually important features. This interaction effect cannot be detected by filter-based methods, justifying the wrapper approach used here [6].

F. Convergence Analysis

Fig. 3 shows clearly different convergence patterns. GA converged over 18 generations with steady improvement, characteristic of evolutionary selection. PSO converged in just 13 iterations, the fastest, but this rapid convergence may have limited search space exploration on the more complex dataset. SA required all 800 iterations, with fitness fluctuating throughout due to probabilistic acceptance, while the global best improved gradually in a stepwise manner.

G. Security Trade-offs and Practical Implications

From an operational cybersecurity perspective, the choice of algorithm depends on deployment context. SA is recommended for high-security environments where minimizing false positives is critical for its FPR of 0.1956% means approximately 2 false alarms per 1,000 benign connections, reducing alert fatigue. GA is preferred for resource-constrained environments such as edge or IoT devices, where its 9-feature model enables faster packet inspection. PSO, despite weaker performance here, suits scenarios requiring frequent retraining due to its 66.7-minute runtime.

The fitness function ($\text{fitness} = 0.9 \times \text{F1} - 0.1 \times \text{features}/78$) explicitly encodes this trade-off by weighting classification accuracy nine times more heavily than feature reduction, reflecting real-world IDS deployment priorities.

H. Limitations

This study has several limitations. First, the training set was capped at 400,000 samples for computational feasibility; future work should be validated on the full uncapped dataset. Second, binary classification was used; extending to multi-class attack detection would provide more granular evaluation. Third, metaheuristic parameters were set based on prior literature and not exhaustive tuning. Parameter sensitivity analysis could yield further improvements. Finally, only Random Forest was used as the base classifier; testing XGBoost or SVM would provide broader insights.

V. CONCLUSION

This paper presented a comparative study of three metaheuristic algorithms, GA, PSO, and SA, for joint feature selection and hyperparameter optimization of a Random Forest-based IDS on the full four-day CICIDS2017 dataset.

SA achieved the best overall results with 99.5879% F1 and 0.1956% FPR using 26 features, demonstrating that temperature-based single-solution exploration is particularly effective on complex, multi-day datasets with diverse attack types. GA achieved the most aggressive feature reduction (88.5%, 9 features) while maintaining competitive performance, making it suitable for resource-constrained deployments. PSO, despite being the fastest to optimize, showed the weakest performance on the full dataset, highlighting its sensitivity to multimodal search spaces.

These findings confirm that metaheuristic optimization is a practical and effective approach to improving IDS performance in realistic network environments. Future work will explore multi-class classification, uncapped training data, and parameter tuning for each algorithm.

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